

## Research Methodology in ELT(ENG518)

### **What is research?**

- Research is a systematic process of investigating a topic or problem.
- It aims to discover new knowledge, validate existing ideas, or solve practical issues.
- Researchers gather data, analyze it, and draw conclusions.

### **General characteristics of research**

- **Rigorous:** Research follows a structured approach with clear steps.
- **Objective:** It seeks unbiased information.
- **Replicable:** Others can repeat the study to verify results.

### **Functions of research**

- **Advancing knowledge:** Research contributes to our understanding of ELT.
- **Improving practice:** It informs teaching methods and materials.
- **Solving problems:** Research addresses real-world challenges.

### **Specific characteristics of ELT research**

- Focuses on language teaching and learning.
- Investigates language acquisition, classroom strategies, and learner needs.

### **Research in ELT (AL)**

- ELT research explores language teaching theories and practices.
- It examines language skills, assessment, and teacher development.

### **What is ELT research about?/Why is research in ELT important?**

- Researchers identify gaps and areas for investigation.
- They ask relevant questions to guide their studies.

### **Demythologizing research**

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- Demystifies complex research concepts.
- Makes research accessible to educators.

## Identifying important questions in ELT

- Researchers identify gaps and areas for investigation.
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## Classification of research

- **Descriptive:** Describes phenomena (e.g., language use in classrooms).
- **Experimental:** Tests hypotheses (e.g., effectiveness of a teaching method).

## Major types of research

- **Qualitative:** Focuses on understanding context and meaning.
- **Quantitative:** Involves numerical data and statistical analysis.

## Educational research

- ELT research falls under educational research.
- It informs educational policies and practices.

## Scientific ways to solve problems

- Scientists use the scientific method to solve problems.
- It involves steps like making observations, asking questions, forming hypotheses, testing predictions, and iterating.
- The goal is to find evidence-based solutions.

## Characteristics of an investigator

- Good investigators possess traits like curiosity, analytical thinking, and integrity.
- They ask the right questions, avoid groupthink, and cope effectively with challenges.

## Assortment of a problem

- Assortment optimization is a common challenge in fields like retail, airlines, and healthcare.
- It involves selecting the right mix of items to offer (e.g., products in a store) to maximize an objective (like revenue).

## Identification of a problem

- Problem identification is crucial.
- It pinpoints the root cause of a dilemma and helps create a clear problem statement.

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- Researchers use it to focus their investigations.

## **Criteria for the selection of a problem**

- A good research problem should be novel, significant, and well-defined.
- It must address a real obstacle and contribute original insights.
- Researchers should ask the right questions to tackle relevant problems.

## **Defining a Problem and Formulating a Hypothesis:**

- A research problem sets the stage for investigation.
- A hypothesis is a testable explanation related to the problem.
- It guides research design and data collection.

## **Kinds of a hypothesis and Characteristics of a good hypothesis**

- Hypotheses can be null (no effect) or alternative (predicting an effect).
- A good hypothesis is specific, testable, and based on existing knowledge.

## **Variables in a hypothesis**

- Variables are factors that change in an experiment.
- Independent variables are manipulated, and dependent variables are measured.

## **How to locate research?**

Researchers use various sources:

- **Preliminary sources:** Gather background information.
- **Secondary sources:** Explore existing research.

## **Locating primary research: exploring databases**

- Primary research involves collecting original data for a specific study.
- To find primary sources, explore databases (like Google Scholar, PubMed, or Scopus).
- Look for research articles, interviews, surveys, or experiments conducted by researchers themselves.

## **LPR: Position papers vs primary research**

- Position papers express opinions or stances on a topic.
- Primary research provides direct evidence (e.g., interviews, data) for analysis.

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- Position papers are secondary sources, while primary research is the raw material.

## **LPR: Tables of references and bibliographies**

- A reference list (bibliography) appears at the end of a research paper.
- It includes all cited sources (primary and secondary).
- Tables of references organize these sources for easy access.

## **LPR: Difference between primary & secondary sources**

- Primary sources offer firsthand information (e.g., original documents, data).
- Secondary sources interpret or summarize primary material (e.g., reviews, articles).
- Both are essential for robust research.

## **Obtaining ELT related research articles**

- Explore ELT (English Language Teaching) journals, databases, and repositories.
- Look for articles on language acquisition, teaching methods, and learner needs.

## **Journals related to ELT research**

- ELT journals publish research findings.
- Examples: "TESOL Quarterly," "Applied Linguistics," and "Language Learning."

## **Exploring ELT research journals**

- Dive into journals to find primary research articles.
- Learn from experts and stay updated on language teaching trends.

## **Sampling terminology**

- Sampling involves selecting a subset from a larger population.
- Terms: sample, population, random sampling, and stratified sampling.

## **Sampling paradigms**

- **Information-Rich Paradigm:** Focuses on detailed data from a few cases.
- **Representative Sampling Paradigm:** Aims for a broader, balanced sample.

## **Ethics in sampling of human participants**

- Researchers must consider ethical guidelines when collecting data.

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- Protect participants' rights, privacy, and well-being.

## Research planning and sampling

- Plan your research design and sampling strategy.
- Ensure your sample represents the larger group accurately.

## Meaning and Definition of sampling

- Sampling is the process of selecting a smaller group (sample) from a larger population.
- It allows researchers to study a manageable subset.

## Sampling techniques and designs

- **Techniques:** random, convenience, purposive, etc.
- **Designs:** cross-sectional, longitudinal, or case studies.

## Characteristics of good sampling

- Representativeness, reliability, and validity matter.
- Avoid sampling errors (e.g., bias, underrepresentation).

## Avoiding errors in sampling

- Be aware of common pitfalls (sampling bias, nonresponse bias).
- Use proper techniques to minimize errors.
- Consider sample size (larger is often better).
- Assess how well your findings apply beyond your sample.

## Classifying research designs

- Researchers use different strategies to structure their studies.

Three continua help classify research designs:

- **Basic-Applied Continuum:** Ranges from fundamental research (understanding principles) to practical research (solving real-world problems).
- **Qualitative-Quantitative Continuum:** Qualitative research (exploring subjective experiences) vs. quantitative research (measuring variables objectively).
- **Exploratory-Confirmatory Continuum:** Exploratory research (generating hypotheses) vs. confirmatory research (testing existing hypotheses).

## Questions and research designs

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## WHAT Questions:

- **Exploratory:** Used to explore new ideas or phenomena.
- **Examples:** Investigating online teaching strategies or student motivation.

## WHY Questions:

- **Confirmatory:** Aim to test existing hypotheses or relationships.
- **Examples:** Studying the impact of a new teaching method or stress on productivity.

## Q&RD: Extraneous factors to avoid

- Researchers must control external influences (e.g., history, maturation) that could affect study outcomes.
- Avoid contamination in control groups and other potential biases.

## EFTA: avoiding various factors

- Researchers must control external influences (e.g., history, maturation) that could affect study outcomes.
- Avoid contamination in control groups and other potential biases.

## Data collection procedures

### Observational Procedures:

- Researchers observe behavior without intervention.
- **Examples:** Watching classroom interactions during teaching experiments.

### Instrumental Procedures:

- Researchers use tools or instruments to collect data.
- **Examples:** Administering surveys or using specialized software.

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## Qualities of a good data collection procedures

- **Reliability Issues:** Reliability refers to the consistency and stability of data collection methods. It ensures that if the same methods are used under the same conditions, they yield consistent results. For example, if you measure the temperature of a liquid sample multiple times using the same thermometer, reliable results would show the same temperature each time.
- **Validity Issues:** Validity relates to how accurately a method measures what it is intended to measure. High validity means that the method produces results that correspond to real properties or characteristics. Ensuring validity is crucial to avoid measuring something different from what you intend. For instance, using a standardized questionnaire based on established theory ensures validity in collecting data on personality traits

## Understanding research results

- **Numerical vs. Verbal Data:** Research results can be presented in numerical (quantitative) or verbal (qualitative) forms. Numerical data involves numbers and graphs, analyzed using statistical methods. Verbal data is expressed in words and analyzed through interpretations and categorizations.
- **Presentation and Analysis:** After collecting data, researchers present and analyze it. This involves organizing findings, creating visual representations (such as charts or graphs), and drawing conclusions.
- **Evaluating Data Quality:** Researchers assess the quality of data by considering factors like accuracy, completeness, reliability, relevance, and timeliness. Ensuring data quality is essential for valid research.
- **Evaluating Explanations and Conclusions:** Researchers critically evaluate their findings, explanations, and conclusions. They consider whether the results align with existing theories and other measures of the same concept.
- **Presentation and Analysis of Numerical Data:** This step involves summarizing numerical data using descriptive statistics (e.g., mean, median, standard deviation) and visualizing it through graphs or charts.

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- **Understanding Statistics of Data:** Researchers interpret statistical measures and understand their implications. This includes understanding concepts like correlation, significance, and confidence intervals.
- **Understanding Statistical Procedures:** Researchers apply statistical techniques (e.g., regression analysis, t-tests) to explore relationships, test hypotheses, and draw meaningful insights from the data.

**Follow the Applied Linguistics Mentor channel on WhatsApp for all the basic help:**

<https://whatsapp.com/channel/0029VaazKAVBfxoDMH4l9U0u>

**Maha Malik**

<https://whatsapp.com/channel/0029VaLvft01XquQQ7sIB12s>

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